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Preplanned Versus Online Control in Baseball Batting: Effects of Temporal Constraints and Spatial Uncertainty

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How does the sensorimotor system control ultrarapid interceptive actions that must be completed within a few hundred milliseconds under high temporal pressure and spatial uncertainty? Such actions, as exemplified by baseball batting, require sophisticated coordination of perceptual and motor processes. The present experiment examined how these ultrafast interceptive movements are controlled by means of predictive and online processes as temporal constraints and spatial uncertainty were systematically varied. Eighteen skilled collegiate batters performed a high-fidelity virtual reality batting task in which time-to-contact (TTC: 500–375 ms) and spatial uncertainty (high/low) were orthogonally manipulated. Hitting performance (contact rate) remained stable at longer TTCs but declined once the available time fell below ∼425–400 ms, identifying a temporal boundary. As temporal urgency increased, swing initiation occurred earlier while swing duration remained constant, indicating predictive timing adjustments. However, timing error increased sharply near this boundary. Nevertheless, bat-head trajectories continued to diverge according to pitch location, and trial-by-trial variability increased at shorter TTCs, revealing within-swing online adjustments. Together, these findings demonstrate that the sensorimotor system coordinates predictive planning and online visual–motor adjustment within a narrow temporal window, suggesting that predictive and online control are integrated in a hybrid manner in temporal-urgency interception.

Public Significance Statement

How does the brain control fast, precise actions when only a few hundred milliseconds are available to respond? Baseball batting provides a clear example of this challenge. Using a high-fidelity virtual reality batting task, this study examined how skilled players balance predictive and online control when facing pitches of varying speed and spatial uncertainty. The findings revealed a critical time limit of around 400 ms, below which hitting performance deteriorates. Skilled batters prepare their swings within this brief time window while still making moment-to-moment corrections during execution. These results show how the sensorimotor system integrates prediction and feedback under severe time pressure and suggest that training near this time limit could improve adaptive coordination and decision-making in fast, dynamic sports and other time-critical activities.

Keywords: interception, baseball, temporal constraints, uncertainty, online control

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All anonymized raw and processed data, analysis code, and experimental materials underlying the findings of this study are publicly available at the Open Science Framework repository: https://osf.io/yz84v/. We have no conflicts of interest to declare. This research was approved by the Ethical Review Committee of the National Institute of Fitness and Sports in Kanoya (Approval 24-1-33).

Hiroki Hayashi served as lead for data curation, formal analysis, investigation, and visualization and contributed equally to conceptualization, methodology, software, validation, and writing–original draft. Sachi Ikudome contributed equally to methodology, validation, and writing–original draft and

continued

When hitting a baseball with a bat, success depends on moving the effector (i.e., the bat) to the right place at the right time to contact the moving object (i.e., the baseball). This example illustrates a broader class of goal-directed actions known as interception, in which an effector (e.g., hand, foot, body, or tool) is controlled to establish a task-relevant spatiotemporal relation with a moving object (e.g., contacting, deflecting, capturing, blocking, or avoiding it). Across species, such interception underpins behaviors ranging from predator–prey pursuit and capture to human everyday activities (e.g., catching an object, shaking hands, collision avoidance), which typically unfold under relatively moderate temporal and informational demands. By contrast, in sports we focus on a distinct form—high-temporal-urgency interception—in which fast-moving objects sharply compress the time available for perception, decision-making, and movement, whereas elevated uncertainty increases informational and internal processing demands; as the available time shrinks and the required time grows, temporal urgency intensifies and imposes qualitatively different demands on the sensorimotor system.

Ball-striking sports such as baseball and cricket illustrate this clearly: A pitched ball can reach the batter in well under half a second, with considerable variability in both temporal (time-to-contact [TTC]) and spatial (location) dimensions. Under such conditions, batters must operate within very narrow temporal and spatial error margins, on the order of tens of milliseconds and a few centimeters, to make solid contact and increase the probability of a successful strike (Kidokoro et al., 2019; Nakashima et al., 2025; Nathan, 2000). These demands are particularly severe when considering that even fast automatic responses to visual stimuli require 110–200 ms (e.g., Brenner & Smeets, 1997; Brenner et al., 1998; Day & Lyon, 2000), and that response times further increase with uncertainty, reflecting a fundamental property of sensorimotor processing: As informational load increases, response times inevitably lengthen (Hick, 1952; Hyman, 1953; Slater-Hammel & Stumpner, 1951). Even after the initiation of movement, the swing requires at least 130 ms (Inkster et al., 2011; Katsumata, 2007; Nicholls et al., 2003). That only 8%–9% of all batted balls across the league in the 2025 Major League Baseball season qualified as "barrels" (a metric for optimal hits) further illustrates the difficulty of performing under such extreme conditions (Baseball Savant, n.d.). These observations raise a fundamental question: How does the sensorimotor system coordinate interception when time for internal processing and action is severely limited by the speed of object motion and uncertainty?

A variety of models have been proposed to explain how interception is controlled across domains such as locomotion, manual reaching, and object tracking (e.g., Beyeler et al., 2023; Casanova et al., 2022; Diaz et al., 2009; Montagne et al., 2004; Scaleia et al., 2015; Tresilian, 2005; Zhang et al., 2025; Zhao & Warren, 2017). Although these models are grounded in distinct theoretical frameworks and characterized by different assumptions and mechanisms (see Wilson et al., 2024), their underlying control principles are often discussed in terms of two broad categories: online (prospective) and model-based (predictive) control (e.g., Arthur et al., 2024; Gray, 2021; Katsumata & Russell, 2012; Wilson et al., 2024; Zhao & Warren, 2015). In this context, online control is characterized by a direct and tight coupling between perception and action, whereby interceptive movements are continuously shaped in real time through the pickup of specifying information embedded in the environment, such as tau (Lee, 1998; Savelsbergh et al., 1991; Tresilian, 1999) and bearing angle (Bastin et al., 2006; Fajen & Warren, 2007; McBeath et al., 1995; McLeod et al., 2006), which are thought to directly determine action-relevant parameters (i.e., when and where to move; Bootsma & van Wieringen, 1990; Gibson, 1979; Lee, 1976). Alternatively, model-based control assumes that movements are guided by predictions (expectations) of future object and body states, generated through internal models that simulate the physical laws of the environment and bodily dynamics (e.g., Clark, 2013; Harris et al., 2022; Nijhawan, 2008; Wolpert & Flanagan, 2001; Wolpert & Ghahramani, 2000; Yarrow et al., 2009; Zago et al., 2004, 2005, 2009). Within this framework, motor plans specify the intended action in advance based on predictions of future states. In this way, key movement parameters (e.g., duration, force, and timing) are predetermined through predictive computations that generate the motor commands classically attributed to a motor program (Kawato, 1999; Keele, 1968; Schmidt, 1975; Wolpert & Kawato, 1998). The resulting actions are executed largely without reliance on immediate sensory feedback during movement (see also Keele, 1981; Summers & Anson, 2009). Thus, to overcome extreme temporal and informational constraints, interception can be accomplished either by acquiring control laws that tightly couple current sensory input to ongoing movement via visual variables, as emphasized in the online control framework, or by forming and relying on precise predictions derived from internal models to compensate for the limited availability of sensory feedback during fast actions, as posited by the model-based control framework.

Both frameworks have received empirical support and contributed to advancing our understanding of the mechanisms underlying high-temporal-urgency interception in sports (e.g., Fajen, 2007; Gray, 2021; Harris et al., 2022, 2023; Müller & Abernethy, 2012; Runswick et al., 2020). Nevertheless, they remain the subject of ongoing debate, stemming in part from the contrasting assumptions they make about how perception guides interception, particularly regarding whether movements are strictly prespecified or can be continuously adjusted during their execution. Historically, rapid interceptive actions such as bat swings were predominantly regarded as preprogrammed, open-loop movements executed with minimal reliance on sensory feedback (e.g., Schmidt, 1975; Schmidt & Lee, 2011; Tyldesley & Whiting, 1975). Consistent with this view, skilled athletes are known to

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rely on early anticipatory cues, such as situational probability (e.g., Abernethy et al., 2001; Cañal-Bruland et al., 2015; Farrow & Reid, 2012; Mann et al., 2014; Murphy et al., 2016; Paull & Glencross, 1997; Runswick et al., 2018), early kinematic information from opponents (e.g., Abernethy & Zawi, 2007; Chen et al., 2017; Jackson & Mogan, 2007; Müller et al., 2006; Savelsbergh et al., 2002; Takamido et al., 2021, 2022; Williams & Davids, 1998), and early ball trajectory (e.g., Gray, 2010; Higuchi et al., 2016), to estimate when and where the ball will arrive and to guide their actions through the weighted integration of contextual, kinematic, and visual information based on their perceived reliability (e.g., Gray & Cañal-Bruland, 2018; Gredin et al., 2018, 2020; Harris et al., 2023; Helm et al., 2020; Nakamoto et al., 2022).

Further evidence for this reliance comes from findings that batters often cease tracking the ball visually around the time of swing initiation and instead shift their gaze toward the anticipated point of contact, where it typically remains throughout the swing (Kishita et al., 2020a, 2020b; Mann et al., 2013; Toole & Fogt, 2021). This suggests that batters may not obtain additional visual information that is functionally relevant for online correction once the swing is underway, reinforcing the notion that the movement is guided in a prespecified manner rather than being continuously corrected during execution. It should be noted that, even when batters cease foveal tracking, motion-related information could in principle still be processed via peripheral vision. However, recent evidence suggests that such peripheral information does not provide an additional advantage for estimating TTC or detecting speed changes following predictive saccades (Vater & Mann, 2023). Consistent with this interpretation, spatial contact accuracy has been reported to remain unaffected in some visual-occlusion paradigms that remove the final portion of the ball's flight (Higuchi et al., 2016), suggesting that late visual information may not be utilized during movement execution (see also Müller et al., 2024). Taken together, these findings have contributed to an increasingly dominant view that rapid interceptive actions in striking sports are governed by predictive control, which relies on early sensory information and is executed without continuous correction during interceptive action.

However, more recently, this view has been challenged by Gray (2021), who argues that it underestimates the extent to which perceptual information continues to shape movement during interceptive action. Building on prior work by Katsumata (2007), who reported a progressive reduction in timing variability across swing phases in baseball batting (landing, weighting, swing, and impact), Gray (2021) further identified strong correlations between adjacent phases, suggesting that compensatory adjustments (i.e., functional coupling) occur as the swing unfolds. For example, if the landing of the front foot occurs slightly later than usual, the subsequent weight shift and initiation of the swing are shortened accordingly, so that the overall timing of bat–ball contact is preserved. Further support for online control comes from Takamido et al. (2025), who applied neural Granger causality analysis of pitcher–batter pairs and demonstrated dynamic coupling between the pitcher's throwing kinematics and the batter's movement execution. Specifically, the pitcher's throwing arm (elbow and wrist) exerted strong causal influence on the batter's wrists and bat tip, whereas lower-body movements such as the back knee and heel also constrained the batter's preparatory actions. This continuous coupling indicates how batters adapt their swing mechanics in real time to the opponent's motion (see also Takamido et al., 2023), supporting the view that movement execution involves online functional regulation rather than rigid preprogramming.

More direct evidence comes from a recent study by Saijo et al. (2025), who investigated baseball swing adjustments under conditions where the ball's trajectory was visually occluded midflight (150 ms after ball release). They found that, compared with the full-vision condition, trajectory adjustments, particularly in the horizontal plane, were significantly impaired under occlusion. Based on this, they proposed that fine-grained control during the swing may rely on continuous visual feedback in the later stages of the pitch (see also Benson et al., 2025). However, because the occlusion occurred approximately 150 ms after release and likely before the onset of most swings, the superior performance under full-vision conditions may also reflect enhanced predictive accuracy afforded by a longer duration of visual information rather than true within-swing online corrections. Nevertheless, the study provides valuable evidence suggesting that continuous visual information contributes to swing regulation.

As outlined earlier, there is still no definitive answer as to which control mode governs high-temporal-urgency interception, largely because existing evidence on bat swing execution remains limited and insufficient to adjudicate competing theoretical claims. First, few studies have tested performance under the severe temporal constraints that batters face in real games. Whereas baseball batters routinely contend with pitches traveling at 150–160 km/hr, most experimental studies have used substantially slower ball speeds of 110–130 km/hr (e.g., Katsumata, 2007; Kidokoro et al., 2019, 2020; Kishita et al., 2020a; Ranganathan & Carlton, 2007), and even the fastest laboratory conditions have reached only approximately 140 km/hr (e.g., Gray, 2002, 2020; Higuchi et al., 2016; Kishita et al., 2020b; Saijo et al., 2025). Laboratory paradigms simulating baseball-like reaching tasks suggest that the control mode of interceptive actions may change as a function of temporal constraints (Ijiri et al., 2015), and that performance can collapse nonlinearly once a critical threshold is exceeded (Kobayashi & Kimura, 2022). These findings underscore the task dependency of control, yet the absence of experiments reproducing the extreme time pressures of real play leaves a decisive gap. Second, systematic manipulations of uncertainty are also lacking. Although some studies have examined the role of predictability (e.g., Gray, 2002; Kidokoro et al., 2020), very few have directly tested how different levels of uncertainty influence swing control, despite evidence from general visuomotor interception tasks that uncertainty can fundamentally alter control modes (Márquez et al., 2024; Márquez & Treviño, 2024). Third, evidence supporting preprogrammed control frequently relies on visual-occlusion paradigms, yet such designs necessarily remove the visual information required for online adjustment (see Cañal-Bruland & Mann, 2025; Müller & Abernethy, 2006). Consequently, these paradigms do not directly test whether online corrections would emerge under naturalistic viewing conditions. Prior work shows that even brief occlusions can impair manual interception (Brenner et al., 2014; Brenner & Smeets, 2011) and increase reliance on predictive strategies based on early trajectory information (Zhao & Warren, 2015). Thus, the absence of observable online corrections in these studies may reflect methodological constraints rather than intrinsic limits of the motor system. Taken together, these limitations indicate that current knowledge remains insufficient to determine how interception is controlled under the most demanding conditions. Addressing this gap by systematically manipulating pitch speed and uncertainty under naturalistic viewing conditions is therefore essential for clarifying the control mechanisms of high-temporal-urgency interception.

The present study aimed to characterize how skilled batters respond to increasing temporal urgency, defined as arising from externally imposed time constraints (shorter TTCs) and internally driven processing demands due to elevated spatial uncertainty, by examining how performance and control processes reorganize or break down under high temporal urgency. To this end, we employed a representative yet controlled virtual reality (VR) batting task that preserved key perceptual features such as full ball flight and required real batting swings, thereby maintaining tight perception–action coupling. External time constraints were manipulated by blocking time-to-contact at six levels ranging from 500 to 375 ms, whereas internal processing demands were varied through high- and low-uncertainty pitch-location conditions. Performance was assessed using complementary measures capturing contact success, bat kinematics, swing timing, and perceptual decision-making.

To evaluate an online-control–dominant account, we formulated the following hypotheses. If continuous online control predominates, contact performance should decline progressively rather than collapse at a fixed boundary as urgency increases. Swing trajectories should continue to diverge according to pitch location during execution, even under severe time constraints. The timing of the swing should show compensatory adjustments, such that later initiation is offset by shorter execution to preserve impact timing. In addition, we assessed perceptual sensitivity to determine whether any performance breakdown reflected perceptual limitations rather than constraints in motor control. Perceptual sensitivity may decrease under increased urgency due to reduced time available for evidence accumulation.

Most prior studies that have argued for online regulation in batting have focused on relatively long-time windows, such as stepping or weight transfer, where adjustments are more intuitively plausible (Gray, 2020; Katsumata, 2007; Müller & Abernethy, 2006). In contrast, the present study isolates the bat swing itself, a rapid phase with only a few hundred milliseconds of duration, to test whether online regulation can be observed even in this very short-latency movement under severe temporal urgency. Establishing such evidence is important for clarifying the true extent of online regulation in batting by determining whether it operates not only in preparatory movements but also within the most time-pressured component of the swing.

Method

Participants

Eighteen male varsity baseball players ($M_{\text{age}} = 21.1 \pm 0.7$ years), all with normal or corrected-to-normal vision, participated in the study. Among them, six were left-handed batters. All belonged to the same university baseball team that regularly competed at the national collegiate level. On average, players had 13.0 ± 1.9 years of formal baseball experience, engaged in regular training sessions for approximately 3 hr per day, 6 days per week, and routinely participated in official intercollegiate matches. Inclusion was limited to players with at least 10 years of continuous baseball experience, active participation in university-level competitions, and regular engagement in structured team practice, ensuring a consistently high skill level across participants. Players ranged in competitive status from regular starters with national tournament experience to those who had participated in official games occasionally, but all were verified by the coaching staff as technically proficient and currently active members. All participants were informed of the experimental procedures in advance and consented to participate. The study was approved by the Ethical Review Committee of the affiliated institution, in accordance with the Declaration of Helsinki (Approval No. 24-1-33).

Sample Size Justification

Because the study required expert batters and six laboratory sessions (see the following text), recruitment was limited to eligible varsity players who could commit to the full schedule. Participants were selected from this pool based on predefined inclusion criteria (see the Participants section), resulting in a feasible sample of 18. In such contexts, "few-observers, many-trials" designs are widely used to improve statistical precision. Increasing trials per participant reduces estimation error and can provide adequate power when the expert population is limited (Baker et al., 2021; Miller, 2024). Accordingly, the study was designed with a high number of trials per participant (768 trials each) to maximize precision.

Because sample size was constrained by population and logistical factors, a conventional a priori power analysis was not appropriate (Lakens, 2022). Following methodological recommendations for resource-constrained studies, we therefore conducted a simulation-based sensitivity analysis (Superpower; Lakens & Caldwell, 2021) to estimate the effect-size magnitudes detectable with $N = 18$. Using the 12 cell means from the 2 (Uncertainty) $\times$ 6 (TTC) design, together with the pooled standard deviation and the average within-participant correlation derived from the observed data, we estimated detectable effects for the TTC main effect, the uncertainty main effect, and their interaction. Simulations indicated high sensitivity for the TTC main effect (simulated $\eta_p^2 \approx .52$; power ≈ 1.00), but limited sensitivity for the smaller uncertainty main effect and the Uncertainty $\times$ TTC interaction (simulated $\eta_p^2 \approx .01$–$.02$; power $< .15$). Because our sample size was fixed, this sensitivity analysis is not intended to retroactively justify the sample size, but rather to transparently describe which effect sizes the study could and could not detect as statistically significant given the available expert population.

To further contextualize these findings, we additionally generated power curves in which the hypothetical sample size was varied from $N = 13$ to $N = 200$ (Figure S1 in the supplemental materials). These curves showed that power for uncertainty-related effects increased only modestly even with substantially larger samples, indicating that limited detectability was primarily driven by small effect sizes rather than sample size alone. Accordingly, the absence of strong uncertainty-related effects in our results may reflect the relatively small influence of uncertainty under the present task conditions.

Transparency and Openness

This research followed the APA Journal Article Reporting Standards for Quantitative Research (Appelbaum et al., 2018). We report sample size determination, data exclusions, manipulations, and measures. Data were collected in 2024–2025. All anonymized raw and processed data, analysis code, and research materials are publicly available at the Open Science Framework repository: https://osf.io/yz84v/. The study and analysis plan were not preregistered. Analyses were conducted using R (Version 4.4.0) and one-dimensional statistical parametric mapping (SPM1d; Pataky, 2010). All descriptive statistics (means and 95% confidence intervals [CIs]) for the dependent measures are provided in the supplemental materials (Tables S1–S4) to support reproducibility in accordance with the Journal Article Reporting Standards for Quantitative Research and open-science recommendations (Appelbaum et al., 2018).

Task and Apparatus

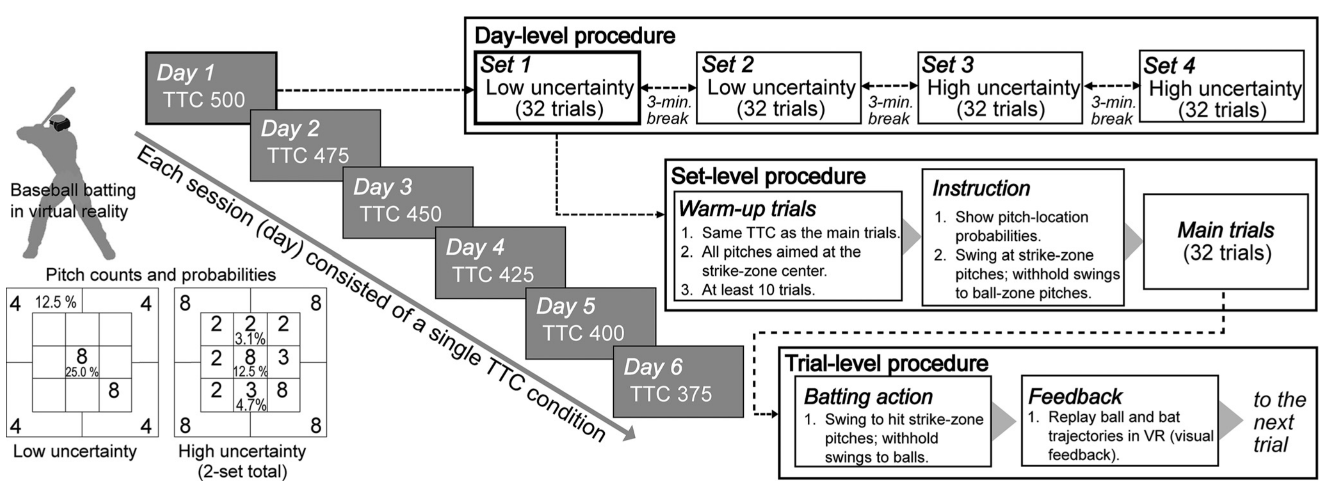
Participants performed a baseball batting task in a VR environment. The task involved two key experimental manipulations: ball speed, operationalized as TTC, and spatial uncertainty, defined as the variability in possible pitch locations. First, TTC was manipulated across six levels, representing the time from ball release to arrival at home plate: 500, 475, 450, 425, 400, and 375 ms. These values corresponded to six ball speeds in 25-ms increments. Second, spatial uncertainty was manipulated by varying the distribution of pitch locations (Figure 1). In the low-uncertainty condition, pitches were restricted to two designated strike-zone locations (center and low-outside corner), whereas in the high-uncertainty condition, pitches were distributed across nine strike-zone locations. Four additional out-of-zone locations were included in both conditions.

The number of pitches per location was weighted so as to reflect patterns commonly observed in collegiate baseball, with a greater concentration at the low-outside corner. However, pitch error is not random but tends to align with the arm-swing plane as a function of arm-slot angle (Shinya et al., 2017). For a right-handed three-quarter arm slot, this produces a diagonally oriented dispersion pattern, such that deviations from a low-outside target tend to extend toward more central regions of the strike zone. Accordingly, a higher probability was assigned to the center location to reflect this directionally structured error distribution. To enable valid comparisons of swing trajectories, the number of pitches to the two designated strike-zone locations was kept equal across all combinations of TTC and uncertainty. For left-handed batters, the pitch layout was mirrored horizontally. In addition, the pitching animation was also mirrored by reversing the *x*-coordinate of the pitcher model, such that a left-handed pitcher was presented in the virtual environment. The strike-zone dimensions were fixed across all participants: Vertically, the zone extended from 0.5 m above the ground (lower bound) to 1.1 m (upper bound); horizontally, it extended from −0.21 m (left edge) to +0.21 m (right edge), centered on home plate. Participants were instructed to respond as they would in actual baseball: If the ball appeared to pass through the strike zone, they were to swing in an attempt to make contact, and if it appeared outside the strike zone, they were to withhold their swing.

A custom virtual baseball environment was developed in Unity (Unity Technologies, San Francisco, CA, United States). The virtual environment depicted a standard baseball stadium with a regulation pitching mound and batter's box. An avatar pitcher delivered the ball at predetermined speeds and locations. The avatar's pitching motion was based on 3D motion capture of a professional pitcher (Eagle System, Motion Analysis Corporation, Santa Rosa, CA, United States). To maintain natural correspondence between pitching motion and ball speed, the avatar's motion was

Figure 1

Overview of the Experimental Design, Pitch-Location Distributions, and Trial Procedure



Note. Each participant completed six sessions conducted on separate days, with each session corresponding to a single time-to-contact (TTC) condition (ranging from 500 to 375 ms). The order of TTC conditions across sessions, the sequence of high- and low-uncertainty blocks within each session, and the presentation order of pitch locations within each set were randomized for each participant. The lower left panels illustrate the pitch-location distributions used to define the low- and high-uncertainty conditions, showing the number of pitches delivered to each location and the corresponding probabilities. Within each session, participants performed four sets of main trials (32 trials per set): two low-uncertainty sets and two high-uncertainty sets, separated by 3-min rest periods. In each trial, participants swung to hit pitches that passed through the strike zone and refrained from swinging at ball-zone pitches; after each trial, ball and bat trajectories were replayed in the virtual environment as visual feedback. VR = virtual reality.

temporally scaled so that playback speed varied proportionally with ball speed. Aside from temporal scaling, pitching kinematics were identical across conditions. To reproduce natural ball trajectories in the virtual environment, a cubic polynomial was calculated to represent the three-dimensional coordinates of the ball over time, following the trajectory calculator proposed by Nathan (2015). To ensure standardization, only a single pitching motion was used for the avatar throughout the experiment, whereas ball trajectories were systematically varied across experimental conditions. This approach was necessary because the experiment included 78 unique pitch variations (13 spatial locations × 6 ball speeds), which would be impractical for a human pitcher to reproduce consistently, particularly with regard to ball speed. Moreover, employing multiple pitchers would have introduced variability in pitching kinematics that could affect participants' batting strategies.

A Valve Index head-mounted display (HMD; Valve Corporation, Bellevue, WA, United States) was used to present an immersive VR environment. The headset features dual 1,440 × 1,600 RGB LCD panels, a 144 Hz refresh rate, ∼0.33 ms low-persistence backlight, and ∼130° field of view. High-precision six-degree-of-freedom tracking of the HMD was achieved using an OptiTrack HMD Clip with Active Tag, a commercially available accessory specifically designed for the Valve Index (NaturalPoint, Inc., Corvallis, OR, United States). The clip contains a preattached active marker array with eight infrared LEDs, enabling robust optical tracking without requiring custom marker placement or calibration. The clip was securely mounted to the front of the headset to ensure stable visibility from multiple camera angles. Motion tracking was performed with 12 Prime x13 cameras (OptiTrack), calibrated using manufacturer-recommended static and dynamic procedures. All cameras operated at 144 Hz, yielding submillimeter spatial accuracy.

Participants used an official baseball bat (85 cm, 890 g) fitted with fourteen 12-mm retro-reflective markers for full six-degree-of-freedom tracking. Positional and orientational data from both the HMD and the bat were streamed in real time to the Unity environment via the OptiTrack NatNet SDK, allowing synchronized visualization of participant actions. This integrated system enabled precise real-time tracking and rendering of visuomotor coordination under immersive and ecologically valid conditions.

Procedure

Each experimental session consisted of a single TTC condition (e.g., 400 ms) and included both high- and low-uncertainty conditions (Figure 1). Because there were six TTC conditions, each participant completed six sessions on separate days. In the high-uncertainty condition, 32 ball-zone and 32 strike-zone pitches were presented (64 trials). In the low-uncertainty condition, 16 ball-zone and 16 strike-zone pitches were presented (32 trials). One set comprised 32 trials. To equate trial numbers, the high-uncertainty block was divided into two sets and the low-uncertainty block was repeated for two sets. Thus, participants completed four sets (128 trials) per TTC condition. TTC order, uncertainty block order, and pitch presentation order were randomized for each participant.

Before the main experiment, participants completed a warm-up and a familiarization phase in the VR environment. In this phase, they wore the HMD and performed a practice block using the same TTC condition as in the main trials, with all pitches aimed at the center of the strike zone. The practice block included at least 10 trials and continued until participants reported feeling comfortable in the VR environment. Before each set, participants were shown a visual representation of the pitch-location probability distribution (based on Figure 1, without specifying pitch counts). This ensured that all participants had equal knowledge of pitch-location probabilities at the beginning of each set, eliminating variability in when they might otherwise become aware of these distributions. Participants were instructed to withhold swings to ball-zone pitches and attempt to hit strike-zone pitches, maximizing successful hits per set. After each trial, both ball and bat trajectories were replayed in the VR environment as visual feedback for the participants, compensating for the absence of haptic feedback in the VR system. Each set lasted ∼10 min with 3-min rest intervals, yielding ∼1-hr sessions. Across the six sessions, each participant completed a total of 768 trials (approximately 6 hr of participation).

Measurements

Dependent measures were computed at the stimulus-condition level (TTC × Uncertainty × Pitch Location). A right-handed coordinate system was defined with the origin at the front edge of home plate: The *z*-axis extended toward the pitcher, the *y*-axis pointed vertically upward, and the *x*-axis was defined as the cross product of y and z. Three-dimensional bat kinematics were recorded at 144 Hz, low-pass filtered with a sixth-order Butterworth filter (15 Hz cutoff) and resampled to 1440 Hz using third-order spline interpolation to provide temporal resolution suitable for detailed analysis of swing dynamics and contact events. The cutoff frequency was selected to attenuate measurement noise while preserving the dominant frequency content of the swing. Raw and processed trajectories were visually inspected to confirm that no spurious distortions were observed. In the high-uncertainty condition, 16 pitches were presented across the two designated strike locations (center and low-outside). In the low-uncertainty condition, only these two locations were used. Analyses were restricted to the first set, yielding 16 pitches across the designated locations and matching the high-uncertainty condition.

Contact Probability

Contact probability served as the primary outcome measure. Contact was defined as any physical collision between the virtual bat and ball in the VR environment, irrespective of contact quality, and thus included both well-struck balls and less optimal contacts (e.g., foul balls). Trials with observed collisions were coded as hits; all others as misses. Contact rate was calculated as the proportion of hits among designated pitches per condition.

Swing Paths and Variability

Mean swing trajectories characterized pitch-specific course control and the timing and extent of trajectory divergence across conditions. Trial-by-trial variability indexed within-swing adjustment capacity, with increased late-swing variability interpreted as potentially reflecting greater reliance on continuous visuomotor regulation. For each participant and condition, bat-head trajectories were segmented from swing onset to 100 ms after crossing the front edge of home plate, time-normalized to 100 samples, and analyzed in the vertical (*Y–Z*) and horizontal (*X–Z*) planes. Mean trajectories were computed at the participant level and averaged across participants. Variability was quantified as the standard deviation of bat-head position at each normalized frame, separately for each designated location.

Temporal Structure of Swing Execution

To examine how batters modulated movement timing under increasing urgency, we quantified swing initiation, duration, and temporal alignment. Swing onset was defined as the first frame in which the resultant velocity of the bat knob exceeded 3 m/s. Duration was measured from onset to when the bat crossed the front edge of home plate. Timing error was the difference between ball and bat arrival times (ball minus bat), with positive values indicating a late bat (Ranganathan & Carlton, 2007).

To evaluate whether compensatory timing strategies during swing execution were reflected in the temporal organization of swing timing, we assessed functional coupling between swing onset and duration. For each TTC condition, swing onset and duration were plotted per trial, and a robust regression line was fitted to each scatter (see Figure 5A). The slope of this regression line was extracted as an index of functional coupling: More negative slopes indicated stronger compensation (i.e., later onsets paired with shorter durations), whereas near-zero slopes reflected diminished temporal adjustment. This analysis does not directly measure online adjustment within a single swing. Rather, following the "good variability" framework (Gray, 2020), the trial-level relationship between swing onset and duration was used as an indirect indicator of whether a compensatory timing signature consistent with within-swing temporal adjustment was present in the organization of swing execution.

Perceptual Discrimination and Response Bias

To dissociate perceptual decision-making from motor execution, a signal detection analysis was conducted on swing/no-swing judgments. Swings to pitches inside the strike zone were treated as hits; swings to pitches outside the zone were treated as false alarms (FAs). Sensitivity (d') and decision criterion (c) were computed from hit and FA rates using standard formulas. Log-linear correction was applied when hit or FA rates reached boundary values (0 or 1), by adding 0.5 to the numerator and 1 to the denominator.

Statistical Analysis

All statistical analyses were conducted using a within-subject (repeated-measures) design. The primary independent variables were TTC (six levels: 500–375 ms in 25-ms steps) and spatial uncertainty (high vs. low). Contact rate was analyzed using a two-way repeated-measures analysis of variance (ANOVA; TTC × Uncertainty).

To assess swing trajectory differences, we used one-dimensional statistical parametric mapping (SPM1d; Pataky, 2010) applied to the time-normalized bat-head trajectories described earlier. SPM1d performs mass-univariate inference across the full trajectory and controls for multiple comparisons using random field theory–based cluster inference. At the group level, paired SPM$\{t\}$ tests were used to compare the two designated strike locations. Trial-by-trial variability was quantified as the standard deviation of bat-head position at each normalized frame, and main effects of TTC and uncertainty were tested with SPM1d. This approach enabled us to identify time intervals in which swing paths diverged and to quantify condition-dependent changes in variability, with increased variability interpreted as evidence of ongoing online visuomotor adjustment.

Swing timing measures (swing initiation time, swing duration, and timing error) were analyzed using separate two-way repeated-measures ANOVAs (TTC × Uncertainty). Post hoc pairwise comparisons were Bonferroni-corrected. Functional coupling between swing initiation time and swing duration was quantified as the slope of the robust regression line described in the Measurements section. To assess whether systematic within-swing temporal compensation was present, we tested whether the slope differed from zero (i.e., no coupling) using one-sample t tests for each TTC condition. In addition, because high- and low-uncertainty trials were pooled within each TTC for slope estimation (Figure 5), slope values were submitted to a one-way repeated-measures ANOVA with TTC as the within-subject factor. Signal detection metrics were analyzed separately. Sensitivity (d') and decision criterion (c), defined and computed as described in the Measurements section, were submitted to two-way repeated-measures ANOVAs (TTC × Uncertainty).

All tests were two-tailed with an alpha level of .05. Effect sizes were reported as generalized eta-squared (η^2_G) for point estimates and partial eta-squared (η^2_p) with 95% CIs. Degrees of freedom were adjusted using the Greenhouse–Geisser correction when sphericity was violated.

Results

For clarity, Table 1 provides a concise summary of the main outcome measures and their associated results across the performance, spatial, temporal, and perceptual domains.

Influence of Time Constraints and Uncertainty on Hitting Performance

Figure 2A provides an overview of the contact rates across TTC and uncertainty conditions. Visual inspection suggests that contact performance remained relatively stable at longer TTCs but dropped sharply once TTC fell below approximately 425 ms. The decline appeared to occur slightly earlier in the high-uncertainty condition, a trend that was partly supported by statistical analyses. A two-way repeated-measures ANOVA with TTC and uncertainty as factors revealed a significant main effect of TTC on contact rate, $F(5, 85) = 16.15$, $p < .001$, $\eta^2_G = .249$; $\eta^2_p = .490$, 95% CI [.32, .59]. In contrast, the main effect of uncertainty was not significant, $F(1, 17) = 0.35$, $p = .563$, $\eta^2_G = .001$; $\eta^2_p = .020$, 95% CI [.00, .27], nor was the TTC × Uncertainty interaction, $F(3.38, 57.46) = 0.59$, $p = .645$, $\eta^2_G = .005$; $\eta^2_p = .030$, 95% CI [.00, .08]. These results indicate that the probability of successful bat–ball contact was primarily constrained by available time, rather than by the level of uncertainty. Post hoc comparisons confirmed that under longer TTC conditions (425–500 ms),

Table 1
Summary of Main Outcome Measures Across Performance, Spatial, Temporal, and Perceptual Domains

Domain	Outcome measures	Results
Performance	Contact rate	• Significant main effect of TTC • Sharp decline at 375–400 ms • No main effect of uncertainty • Under high uncertainty, contact remained higher for designated versus other locations
Spatial regulation	Bat-head trajectories	• Trajectories diverged by pitch location • Earlier and broader separation at longer TTCs
	SD of bat-head position	• Greater late-phase variability at 375–400 ms • Effect primarily observed for center pitches
Temporal regulation	Swing initiation time	• Significant main effect of TTC • Initiation progressively advanced as TTC shortened • No uncertainty effect
	Swing duration	• No significant effects of TTC or uncertainty • Duration remained stable across conditions
	Timing error	• Significant main effect of TTC • Larger errors at 375–400 ms • No uncertainty effect
	Initiation–duration coupling	• Slopes did not differ significantly from zero at any TTC • No significant effect of TTC
Perceptual decision-making	Sensitivity (d')	• Significant main effect of TTC • Gradual decrease as TTC shortened • No uncertainty effect
	Decision criterion (c)	• No significant effects • Values consistently negative (liberal bias)

Note. TTC = time-to-contact.

contact rates were consistently high with no significant differences. In contrast, performance deteriorated markedly under shorter TTCs: At 400 ms, contact rates were significantly lower than at 425, 475, and 500 ms (all ps $\leq$ .05). Furthermore, 375 ms showed an even greater decline compared with all conditions from 425 to 500 ms (all ps < .001). Thus, batting performance was stable when at least 425 ms of available hitting time was present, but fell off in a nonlinear, threshold-like fashion below this range.

Next, to explore why no clear effect of uncertainty was observed, we considered the possibility that hitters might maintain their performance in the high-uncertainty condition by focusing on the designated strike location, which had relatively high pitch probabilities, while sacrificing performance for other zones. To test this, we compared the contact rates for the designated strike locations that were common to both uncertainty conditions with those for the other strike zones that were included only in the high-uncertainty condition (Figure 2B). A two-way repeated-measures ANOVA with TTC and ball course (two levels: designated strike locations vs. other locations) revealed significant main effects of TTC, $F(5, 85) = 21.56$, $p < .001$, $\eta^2_G = .261$; $\eta^2_p = .560$, 95% CI [.41, .65], and course, $F(1, 17) = 29.90$, $p < .001$, $\eta^2_G = .142$; $\eta^2_p = .640$, 95% CI [.31, .79], whereas the TTC × Course interaction was not significant, $F(2.59, 44.08) = 1.21$, $p = .314$, $\eta^2_G = .015$; $\eta^2_p = .070$, 95% CI [.00, .14]. Most importantly, the course effect was pronounced. Contact rates for the designated strike locations were substantially higher than for the other locations (estimate = 16.15, 95% CI [9.92, 22.38], $p < .001$). As expected, contact rates for the other courses were markedly lower than those for the designated zones, supporting the idea that hitters preserved performance in the designated zones even under high uncertainty, while performance in other zones declined substantially.

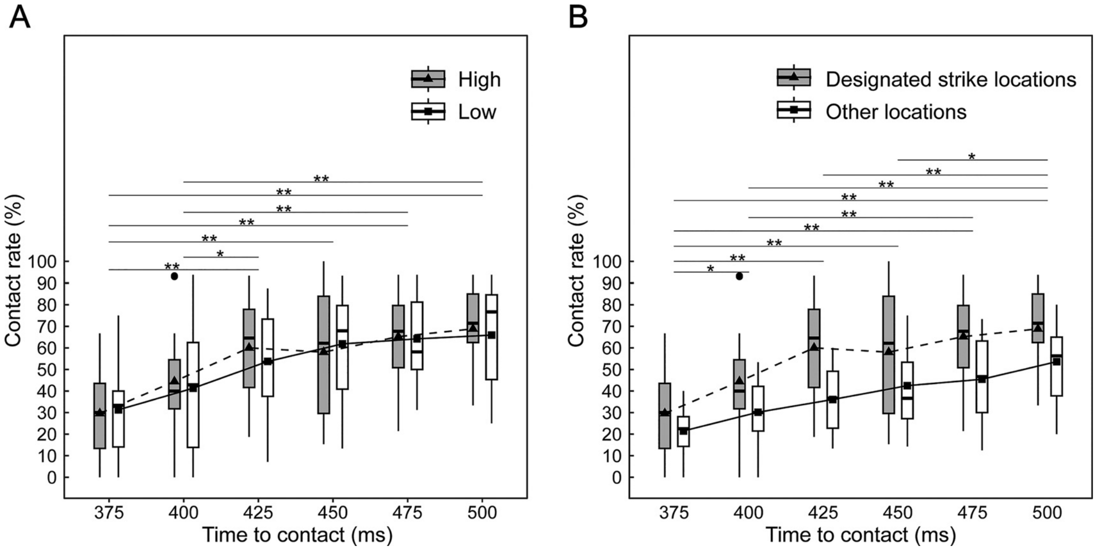
Influence of Time Constraints and Uncertainty on Spatial Regulation

Figure 3 shows the averaged bat-head trajectories for pitches delivered to the center and the low-outside corner of the strike zone across TTC conditions. Because previous analyses of contact rates revealed no effect of uncertainty and bat trajectory patterns were similar between high- and low-uncertainty conditions, data from both uncertainty conditions were merged for the SPM1d analysis. Panels A and B depict trajectories in the vertical (*Y–Z*) and horizontal (*X–Z*) planes, respectively.

Across conditions, swings to the low-outside pitch were directed downward relative to the center pitch in the *Y–Z* plane and extended slightly outward in the *X–Z* plane. SPM1d identified significant intervals during which trajectories to the two pitch locations diverged (shaded regions in the right-hand panels). In the *Y–Z* plane (Figure 3A), divergence was observed at 42%–93% of the swing at 375 ms (interval length 51%), 32%–92% at 400 ms (60%), and 35%–92% at 425 ms (57%). At longer TTCs, the intervals widened (25%–94% at 450 ms, 31%–95% at 475 ms, 21%–95% at 500 ms), and the SPM{t} peaks increased substantially (approximately 15 at 475–500 ms vs. around 10 at shorter TTCs), indicating stronger trajectory separation when more time was available. In the *X–Z* plane (Figure 3B), divergence generally appeared later but showed the same trend. At 375 ms, separation was restricted to 67%–100% (33%) and at 400 ms to 65%–100% (35%), with SPM{t} peaks of approximately 7. The interval broadened at 425 ms (45%–100%, 55%), 450 ms (58%–100%, 42%), 475 ms (58%–100%, 42%), and 500 ms (38%–100%, 62%), accompanied by higher SPM{t} peaks of approximately 12–13 for the longer TTCs. Hence, under shorter TTCs, separation was confined to narrow, late phases of the swing, whereas under longer TTCs, divergence extended over earlier and broader intervals and occurred with greater magnitude.

Figure 2

Contact Rates as a Function of Time-to-Contact (TTC)



Note. Panel A: Contact rates in the high- and low-uncertainty conditions for the designated strike locations (center and low-outside corner). Triangles and squares indicate mean values for the high- and low-uncertainty conditions, respectively. Panel B: Contact Rates for target balls (designated strike locations) and other balls in the high-uncertainty condition. Boxplots represent the median (horizontal line), interquartile range (box), and data range (whiskers). Asterisks indicate significant pairwise differences across TTC conditions (* $p < .05$. ** $p < .01$).

Panels C and D illustrate within-course bat trajectory variability, defined as the variance of bat-head positions across trials for each pitch location. Variability increased progressively throughout the swing and reached its maximum near the contact phase in all conditions. However, the magnitude of variability differed systematically with TTC. At shorter TTCs (375–400 ms), variability rose more steeply and peaked at higher values, indicating greater trial-to-trial dispersion of swing trajectories. In contrast, at longer TTCs (450–500 ms), variability increased more gradually and remained lower at contact, reflecting more consistent execution. SPM analyses further revealed that this TTC-dependent increase in variability was evident for center pitches in both the *Y* (vertical) and *X* (lateral) directions. Yet, for low-outside pitches, no reliable TTC-related modulation of variability was observed in either direction. Thus, the destabilizing effect of shorter TTCs on swing consistency was specific to the center course, whereas variability to the low-outside course remained relatively unaffected by time constraints.

Taken together, these findings indicate that under longer TTCs, batters achieved earlier and more sustained separation of swing trajectories across pitch locations with relatively low variability, whereas shorter TTCs constrained adjustments to late and narrow phases of the swing and amplified trial-to-trial variability, particularly for center pitches.

Influence of Time Constraints and Uncertainty on Temporal Regulation

Temporal Structure

Figure 4 summarizes the temporal structure of swing movements across TTC and uncertainty conditions. The top panels (A) display stacked bar graphs showing the mean swing initiation time, swing execution time, and timing error for each TTC level under high- and low-uncertainty conditions. As TTC shortened, initiation time decreased progressively, swing execution time remained relatively stable, and timing error tended to increase. These temporal components were further examined using separate two-way repeated-measures ANOVAs for each dependent variable (Panels B–D).

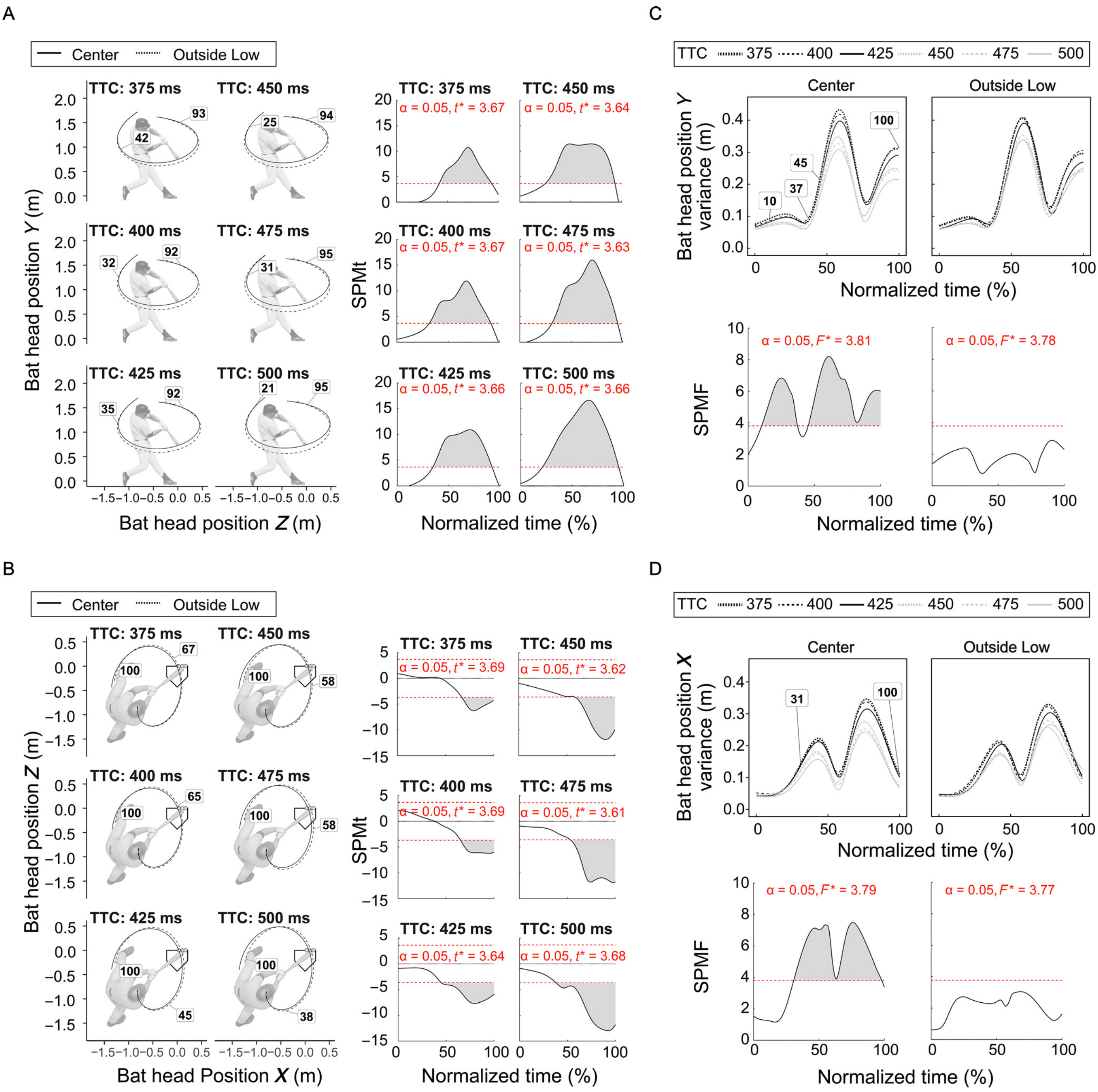
Figure 4B presents swing initiation time. The main effect of TTC was significant, $F(5, 85) = 333.96$, $p < .001$, $\eta^2_G = .840$; $\eta^2_p = .952$, 95% CI [.93, .96], whereas neither the main effect of uncertainty, $F(1, 17) = 4.32$, $p = .053$, $\eta^2_G = .007$; $\eta^2_p = .204$, 95% CI [.00, .50], nor the interaction, $F(5, 85) = 1.18$, $p = .326$, $\eta^2_G = .004$; $\eta^2_p = .064$, 95% CI [.00, .14], reached significance. Post hoc comparisons revealed significant differences between all TTC levels (*p*s < .05), including each adjacent level from 500 ms down to 375 ms. Specifically, initiation time at 500 ms was significantly later than at 475 ms, which was significantly later than at 450 ms, and so on down to 375 ms. This pattern was consistent across both uncertainty conditions, indicating a systematic advancement of swing onset with increasing temporal urgency.

Figure 4C illustrates swing execution time. A two-way repeated-measures ANOVA revealed no significant main effect of TTC, $F(2.98, 50.63) = 0.13$, $p = .944$, $\eta^2_G = .002$; $\eta^2_p = .007$, 95% CI [.00, .00], and no significant main effect of uncertainty, $F(1, 17) = 0.10$, $p = .758$, $\eta^2_G = .000$; $\eta^2_p = .006$, 95% CI [.00, .22]. The interaction was also not significant, $F(5, 85) = 1.04$, $p = .398$, $\eta^2_G = .002$; $\eta^2_p = .061$, 95% CI [.00, .13].

Figure 4D shows timing error. A two-way repeated-measures ANOVA revealed a significant main effect of TTC, $F(5, 85) = 8.04$,

Figure 3

Bat Swing Trajectories and Trial-by-Trial Variability as a Function of Time-to-Contact and Pitch Location

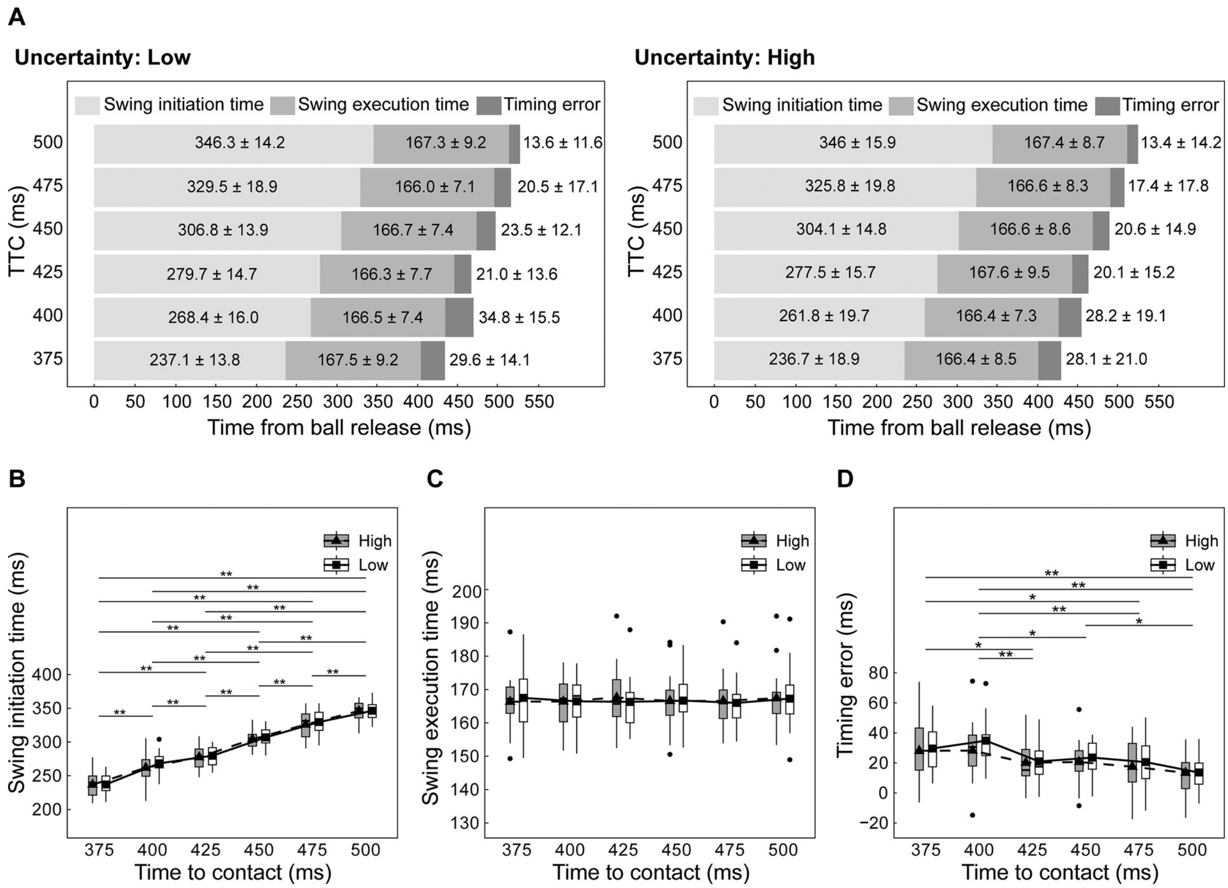


Note. Panel A: Mean bat-head trajectories projected onto the sagittal (*Y*–*Z*) plane for center (solid lines) and low-outside (dashed lines) pitch locations across six TTC conditions (375–500 ms). Panel B: Mean bat-head trajectories projected onto the transverse (*X*–*Z*) plane. Panels C–D: Trial-by-trial variability (standard deviation) of bat-head position, shown separately for the vertical (*Y*; Panel C) and lateral (*X*; Panel D) directions, plotted for center and low-outside pitch locations and color-coded by TTC. For each panel, the corresponding statistical parametric mapping (SPM1d) results are shown in adjacent subpanels, with shaded regions indicating time intervals of significant effects ($p < .05$). Numbers adjacent to the trajectory or variability curves indicate the onset and offset (% of the normalized swing) of these significant intervals. All data are averaged across uncertainty conditions. SPM{t} and SPM{F} indicate the t and F statistics from statistical parametric mapping, respectively. See the online article for the color version of this figure.

$p < .001$, $\eta^2_G = .135$; $\eta^2_p = .320$, 95% CI [.14, .44], whereas the main effect of uncertainty did not reach significance, $F(1, 17) = 3.56$, $p = .076$, $\eta^2_G = .007$; $\eta^2_p = .170$, 95% CI [.00, .47], and the interaction was also not significant, $F(5, 85) = 1.09$, $p = .371$, $\eta^2_G = .005$; $\eta^2_p = .064$, 95% CI [.00, .13]. Post hoc comparisons revealed selective differences in timing errors across TTCs. Specifically, 375 ms produced greater errors than 425, 475, and 500 ms ($p < .01$), whereas 400 ms showed greater errors than 425, 450, 475, and

Figure 4

Swing Timing Components Across Time-to-Contact (TTC) and Spatial Uncertainty Conditions



Note. Panel A: Average temporal structure of batting movements under low (left) and high (right) uncertainty conditions across six TTC levels. Each stacked bar represents the mean swing initiation time (light gray), swing execution time (medium gray), and timing error (dark gray), measured relative to ball release. Panels B–D: Boxplots of swing initiation time (B), swing execution time (C), and timing error (D) for each TTC and uncertainty condition. Boxplots show the median, interquartile range, and whiskers.
Asterisks indicate significant pairwise differences across TTC conditions (* $p < .05$. ** $p < .01$).

500 ms (all *ps* < .05). These patterns indicate that timing errors began to increase noticeably around 400 ms, reflecting the heightened temporal urgency at shorter TTCs. In contrast, no consistent significant differences were found among TTCs from 425–500 ms, with most pairwise comparisons showing $p > .05$, suggesting that performance stabilized once at least 425 ms of hitting time was available.

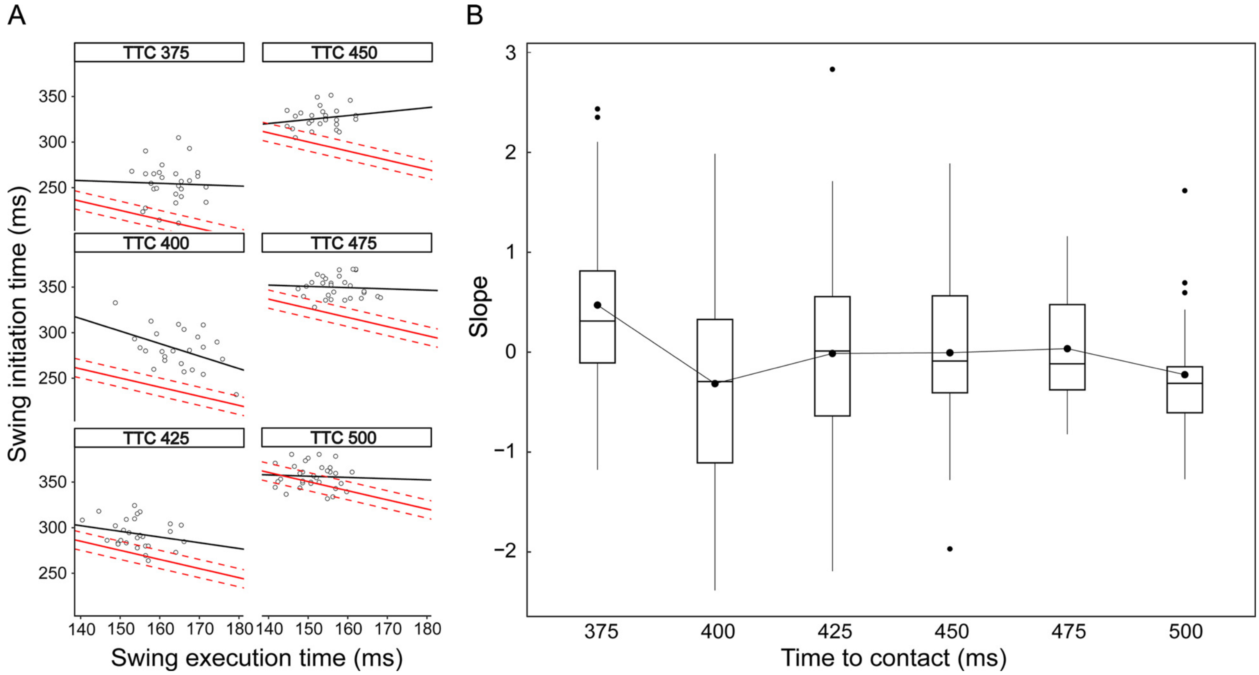
Taken together, these results demonstrate that, similar to contact rate, timing accuracy deteriorated sharply under high temporal urgency (375–400 ms) but stabilized at a relatively consistent level from 425 ms onward. This stabilization was achieved not by lengthening or shortening the swing itself, which remained nearly constant across conditions, but by systematically advancing swing initiation as TTC shortened (see Figures 4B and 4C). However, despite significant stepwise differences across all TTC levels, the magnitude of initiation shifts at 400 ms and 375 ms was insufficient relative to the progressive 25-ms reductions in TTC. For example, initiation time at 400 ms was only about 10 ms earlier than at 425 ms, and initiation time at 375 ms lagged behind the level that would have been required to fully compensate. These incomplete adjustments account for the larger timing errors observed at the shorter TTCs, specifically 400 and 375 ms (Figures 4A and 4D).

Functional Coupling of Swing Timing Parameters

Because each TTC condition was presented in blocks, batters could adopt a simple strategy of shifting initiation time across conditions, as shown in Figure 4. This block-based comparison makes it difficult to detect within-swing timing coordination that may occur within a given TTC condition. To examine whether temporal coordination occurring during swing execution is reflected in the overall timing structure of the swing, we analyzed the relationship between swing initiation time and swing duration within each TTC condition (Figure 5). However, because the temporal structure of batting (swing initiation time, swing time, and timing error) showed no main effect of uncertainty and no TTC × Uncertainty interaction (Figures 4B–4D), high- and low-uncertainty trials were pooled within each TTC level for estimation and reporting. This pooling reduces model complexity and improves estimate precision, given that uncertainty did not modulate the temporal organization of swing timing.

Figure 5

Within-Swing Relationship Between Swing Initiation Time and Swing Execution Time Across Time-to-Contact (TTC) Conditions



Note. Trials from high- and low-uncertainty conditions were pooled within each TTC. Panel A: A representative participant is shown, where each dot represents one swing trial, and the diagonal line indicates combinations of initiation and execution times that would yield zero timing error. For each participant and TTC, a robust regression line was fitted to the distribution of trials, and its slope was extracted. Panel B: Box-and-whisker plots summarize these slope values across participants for each TTC condition. For reference, a slope of −1 would correspond to perfect compensation along the TTC line. See the online article for the color version of this figure.

A representative participant is shown in Figure 5A, where each point denotes a single trial plotted relative to the diagonal line representing combinations of initiation and execution times that yield zero timing error. This visualization illustrates the relationship between initiation and execution times within a TTC condition. To quantify this relationship across participants, we fitted a robust regression line to the distribution of trials for each TTC condition and extracted its slope. Figure 5B summarizes these slopes averaged across participants. For reference, a slope of −1 would correspond to perfect compensation along the TTC line. One-sample t tests revealed that, for all TTC conditions, the mean slope did not differ significantly from zero, with 95% CIs consistently including zero (all ps > .05). Specifically, the mean slope (95% CI) was 0.47 [−0.05, 1.00] at TTC 375, −0.32 [−0.83, 0.20] at 400, −0.01 [−0.59, 0.57] at 425, −0.01 [−0.48, 0.47] at 450, 0.04 [−0.28, 0.35] at 475, and −0.23 [−0.57, 0.12] at 500. Accordingly, no systematic within-swing temporal compensation between swing initiation time and swing duration was detected within any TTC condition. A complementary one-way ANOVA likewise revealed no significant effect of TTC on slope values, $F(5, 85) = 1.75$, $p = .131$, $\eta^2_G = .070$; $\eta^2_p = .090$, 95% CI [.00, .19].

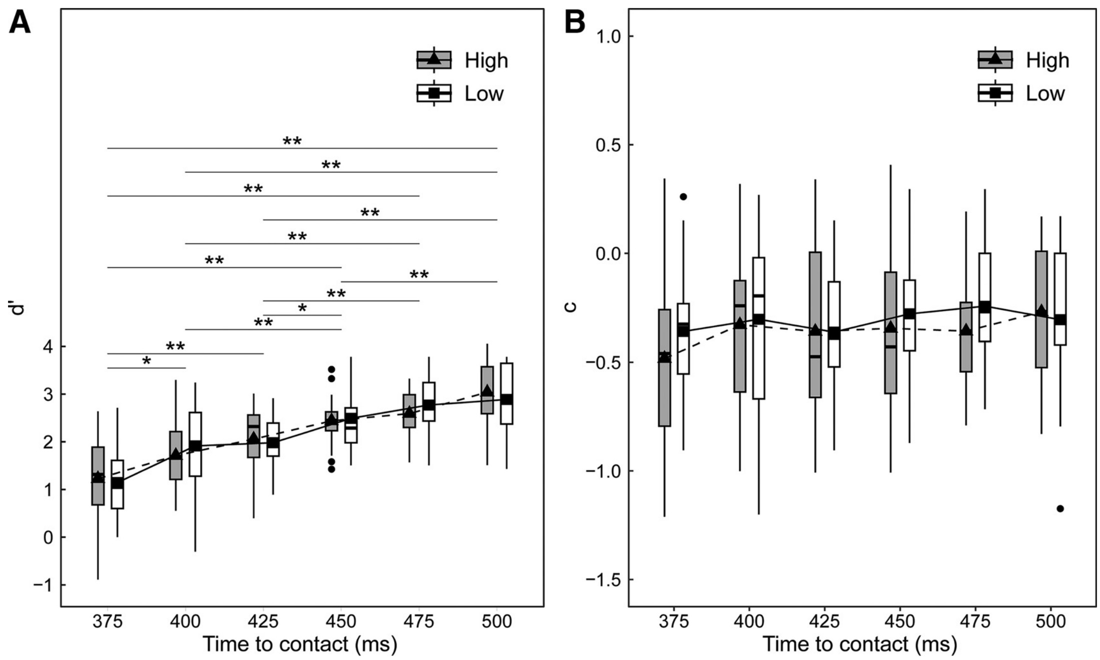
Perceptual Sensitivity (d') and Decision Bias (c)

In line with signal detection theory, perceptual sensitivity (d') reflects the ability to discriminate between strike and ball pitches, with higher values indicating more accurate discrimination independent of response bias. Figure 6A shows d' across TTC and uncertainty conditions. A two-way repeated-measures ANOVA revealed a significant main effect of TTC, $F(5, 85) = 31.44$, $p < .001$, $\eta^2_G = .418$; $\eta^2_p = .650$, 95% CI [.52, .73], whereas neither the main effect of uncertainty, $F(1, 17) = 0.09$, $p = .773$, $\eta^2_G < .001$; $\eta^2_p = .000$, 95% CI [.00, .21], nor the interaction, $F(5, 85) = 0.74$, $p = .596$, $\eta^2_G = .009$; $\eta^2_p = .040$, 95% CI [.00, .10], was significant. Post hoc comparisons showed that d' was markedly reduced at shorter TTCs (375–400 ms) compared with longer TTCs. Specifically, 375 ms was significantly lower than 400–500 ms (all $p < .01$), and 400 ms was lower than 450–500 ms ($p < .01$). Moreover, 425 ms was lower than 450–500 ms ($p < .05$), and 450 ms was still lower than 500 ms ($p = .002$). In contrast, no significant difference was observed between 475 and 500 ms. Taken together, these results demonstrate that perceptual sensitivity declined gradually as temporal constraints became stronger, with the lowest values observed at 375–400 ms. Unlike contact rate or timing error, which showed more threshold-like drops around 425 ms, d' decreased in a graded fashion across TTCs, indicating a stepwise loss of discrimination ability under increasing temporal urgency.

In signal detection theory, decision bias (c) represents the response tendency, indicating whether participants are more inclined to classify pitches as strikes (liberal bias, negative values) or as balls (conservative bias, positive values), independent of discrimination sensitivity. Figure 6B shows the distribution of c across TTC and uncertainty conditions. The following analyses examined how decision bias was influenced by temporal constraints and spatial

Figure 6

Signal Detection Measures as a Function of Time-to-Contact (TTC), Plotted Separately for High- and Low-Uncertainty Conditions



Note. Panel A: Perceptual sensitivity (d'), reflecting participants' ability to discriminate between strikes and balls independent of response bias. Panel B: decision bias (c), indicating participants' response tendency, with negative values reflecting a liberal bias (tendency to classify more pitches as strikes) and positive values reflecting a conservative bias (tendency to classify more pitches as balls).
Asterisks indicate significant pairwise differences across TTC conditions (* $p < .05$. ** $p < .01$).

uncertainty. A two-way repeated-measures ANOVA with TTC and uncertainty as factors revealed no significant main effects and no interaction: TTC, $F(5, 85) = 1.271$, $p = .284$, $\eta_G^2 = .018$; $\eta_p^2 = .070$, 95% CI [.00, .15]; uncertainty, $F(1, 17) = 2.459$, $p = .135$, $\eta_G^2 = .005$; $\eta_p^2 = .013$, 95% CI [.00, .43]; interaction, $F(5, 85) = 0.559$, $p = .701$, $\eta_G^2 = .008$; $\eta_p^2 = .030$, 95% CI [.00, .08]. However, c values were consistently negative across conditions, suggesting that participants maintained a liberal bias (i.e., a greater tendency to call pitches strikes), and this tendency was not modulated by temporal constraints or spatial uncertainty.

Discussion

The present study examined how baseball batters control their swings under conditions of high temporal urgency, addressing the long-standing debate between preplanned and online control in interceptive actions. Three primary findings emerged. First, contact performance declined sharply when effective TTC fell below approximately 425–400 ms, despite stable performance at longer TTCs. This threshold-like pattern was evident in both contact rate (Figure 2A) and timing error (Figure 4D), revealing a temporal boundary at which the combined effects of external time pressure and internal processing delays exceeded the system's control capacity. Second, bat-head trajectories exhibited systematic course-specific divergence between center and low-outside pitches, but this divergence was delayed and spatially compressed under higher temporal urgency (Figures 3A and 3B). When more time was available, trajectories separated earlier and more distinctly, with relatively low trial-by-trial variability. In contrast, at shorter TTCs, divergence occurred later during the swing and variability increased (Figures 3C and 3D). Third, swing initiation time advanced progressively with increasing urgency (Figure 4B), while swing duration remained constant across TTC conditions (Figure 4C). Thus, the available time for movement preparation decreased systematically as TTC shortened. Together, these findings indicate that even within the brief duration of a bat swing, movement kinematics were modulated in response to task constraints, revealing moment-to-moment spatial adjustments under high temporal urgency. However, several aspects of the results appear less consistent with an exclusively online control account and may instead align more closely with control processes that are established in advance of movement.

Evidence for Online Control

Clear behavioral signatures indicate that batting involves continuous online control driven by currently available visual information (Gray, 2021; Zhao & Warren, 2015). A particularly compelling indicator of this is the systematic divergence of bat-head trajectories as a function of pitch location (Figure 3). Although trajectories initially overlapped, they separated between center and low-outside pitches during the swing, and the timing of this separation varied with TTC. When more time was available, divergence occurred earlier and more distinctly. Even under the shortest TTCs, after the swing had already been launched, trajectories still diverged late in the movement, showing that batters continued to modify the bat path while the swing was unfolding. Importantly, swing duration remained approximately constant at 160–170 ms across TTC conditions (Figure 4), indicating that these temporal shifts in divergence were not due to differences in movement duration but reflected genuine online modification. Such delayed yet distinct divergence under high temporal urgency is inconsistent with a purely prespecified plan, which would predict a fixed, TTC-independent trajectory once the movement is initiated (e.g., Tresilian, 2005; Tyldesley & Whiting, 1975). Instead, this temporal shift in divergence indicates that batters use incoming visual information to reshape their motion online, in line with continuous perception–action coupling models of interception (e.g., Bootsma & van Wieringen, 1990; Brenner & Smeets, 2011; Lee, 1976, 1998; Montagne et al., 1999; Savelsbergh et al., 1991; Tresilian, 1999). Additional support for online adjustment is provided by the increase in trial-by-trial variability of swing trajectories observed under shorter TTCs (Figure 3C). Trial-to-trial variability may, at least in part, reflect differences in the strength or frequency of within-trial feedback corrections (e.g., Brenner & Smeets, 2009). In this context, enhanced variability does not signify a loss of control per se but rather reflects the increased recruitment of online adjustment.

Taken together, these findings demonstrate that batting relies on continuous, within-swing adjustments guided by current visual information rather than on a fixed prespecified plan. In doing so, they directly support Gray's (2021) call to reconsider the role of online visual control in batting and challenge long-held assumptions that the swing is governed solely by predictive mechanisms. The present results thus reinforce the view that visual–motor coupling remains operative throughout the swing, even under severe temporal urgency.

Evidence for Preplanned Control

Beyond the evidence for online adjustment, several findings point to a preplanned component of batting control. Classic information-processing (e.g., Schmidt, 1975, 2003) or model-based accounts (e.g., Wolpert & Ghahramani, 2000; Wolpert & Kawato, 1998) propose that key movement parameters are specified in advance of movement onset through predictive computations. In interceptive tasks, this implies that both the temporal and spatial structure of the swing must be organized before execution based on predictions of ball motion and bodily state.

A key behavioral signature of this preplanned organization appeared in the temporal domain. As TTC shortened, batters systematically advanced swing initiation, demonstrating anticipatory adjustment to increasing time pressure. This anticipatory shift was not accompanied by systematic within-swing temporal compensation between swing initiation time and swing duration (Figure 5B). This pattern indicates that the decision of when to start moving was made in advance by integrating the internally represented swing duration with the predicted time of ball arrival (see Brenner & Smeets, 2018). Furthermore, this anticipatory shift reached a saturation around 400 ms, beyond which timing errors increased sharply (Figure 4A). The inability to further advance initiation and the concurrent rise in temporal error indicate that the internal processes required for movement preparation had reached their limit. This breakdown was accompanied by a steep decline in contact rate (Figure 2A), suggesting that once the available time dropped below the functional preparatory boundary—roughly 400 ms encompassing perceptual evaluation, movement planning, and initial execution—the motor system could no longer coordinate the swing with sufficient precision. A similar threshold-like deterioration in performance has been reported in a baseball-like Go/No-go task (Kobayashi & Kimura, 2022), where decision efficacy and hitting accuracy collapsed when TTC fell below approximately 500 ms. Notably, unlike the sharp decline in perceptual sensitivity observed in that study, perceptual sensitivity in the present data decreased only gradually with increasing urgency. This dissociation suggests that perceptual judgment remained relatively intact even as motor performance collapsed, consistent with the interpretation that the breakdown originated not from degraded perception but from insufficient time for motor planning within this functional boundary.

Spatial characteristics of the swing further support this interpretation. When sufficient time was available, swing trajectories exhibited early and stable divergence between pitch courses (Figures 3A and 3B), and trial-by-trial variability remained low (Figure 3C), indicating the use of a consistent, preparameterized movement pattern rather than one shaped by moment-to-moment feedback. Under high uncertainty, spatial planning became more selective: Batters preserved accuracy for high-probability ("designated") pitch locations but showed marked declines for low-probability courses (Figure 2B). If swing execution were governed primarily by continuous visual feedback, comparable accuracy would be expected across all visible pitches within the strike zone. Instead, this selective preservation indicates that batters strategically concentrated their preparatory resources on the most likely pitch locations to maintain performance under uncertainty. Given the limited time available for motor planning, it would be inefficient to distribute preparation evenly across all potential trajectories. The observed pattern therefore reflects a process of anticipatory selection, in which only a subset of probable spatial outcomes is parameterized in advance to minimize cognitive and motor load. Within a predictive framework, such selectivity can be understood as a bias toward expected spatial outcomes under temporal constraints.

Taken together, these findings provide behavioral evidence consistent with model-based and information-processing accounts of motor control, in which key movement parameters are specified in advance of movement onset. The functional preparatory boundary of roughly 400 ms, the selective spatial preconfiguration toward probable pitch locations, and the dissociation between perceptual sensitivity and motor performance collectively indicate that predictive planning constrains the temporal and spatial organization of the swing.

Hybrid Control: Predictive Scaffold and Online Refinement

Although the preceding sections have identified behavioral signatures consistent with both online control and preplanned control, the present findings cannot be reduced to either account alone. Instead, rapid interceptive action appears to emerge from a dynamic interplay between predictive planning and continuous online correction within the functional preparatory boundary described earlier. To understand how these processes coexist, it is useful to refer to frameworks that explicitly describe the integration of prediction and feedback during movement execution. One such account is provided by Brenner and Smeets (2018), who proposed that interceptive movements are planned in advance and then continuously controlled using the latest available information, with continuous control serving not only to correct deviations from the plan but also to adjust the plan itself as new sensory information becomes available (see also Katsumata & Russell, 2012; Müller & Abernethy, 2012). In line with this account, the present data reveal behavioral patterns consistent with a continuous predictive–corrective process. When sufficient time is available, this plan can be formulated with relatively high accuracy and further refined through continuous feedback, leading to well-coordinated and stable trajectories. As temporal urgency increases, however, the initial plan likely becomes less precise, which increases the demand for rapid correction. With shorter TTCs, both preparation time and the opportunity for updating become progressively compressed. Once swing initiation can no longer be advanced to compensate for the reduced time, the same predictive–corrective process must operate within a very narrow temporal margin. Under such conditions, the increasing uncertainty of the plan and the limited time for adjustment may produce high-gain, short-lived corrections that are not clearly expressed in trajectory form but instead appear as greater variability, reflecting the system's attempt to preserve coordination under increasing uncertainty. When TTC falls below approximately 400 ms, where initiation could not be further advanced, control seems to deteriorate abruptly. This pattern may not reflect a shift in control mode but rather the time-limited expression of a continuous predictive–corrective process under increasingly uncertain conditions.

This behavioral signature of continuous adjustment under constraints also aligns with computational models of motor control that emphasize optimal use of sensory feedback. Computationally, this continuous predictive–corrective process can be interpreted within the internal-model framework (Wolpert & Ghahramani, 2000; Wolpert & Kawato, 1998), in which forward models generate sensory predictions that are continuously updated based on incoming sensory feedback and resulting prediction errors. Within this architecture, optimal feedback control (Scott, 2016; Todorov & Jordan, 2002) provides a formal description of how these predictions are used to minimize task error and control cost through selective feedback gains. Under shorter TTCs, the accuracy of forward predictions likely declines, potentially requiring higher feedback gains, which may be expressed behaviorally as increased variability under severe time constraints. At a broader theoretical level of neural computation, these mechanisms align with predictive processing and active inference views (Adams et al., 2013; Friston, 2010; Friston et al., 2016), which generalize internal modeling and feedback optimization into a unified principle of hierarchical prediction-error minimization across time and levels of control. In this way, predictive processing embeds hybrid control within a general principle of brain function: the continuous minimization of prediction error across time and hierarchical levels. This framework has also been extended to account for skilled performance and motor learning in complex sport contexts (Harris et al., 2022, 2023, 2025). Taken together, these converging perspectives suggest that the hybrid control observed in the present study constitutes a behavioral instantiation of this broader computational principle, wherein predictive and feedback processes operate as a unified system to sustain goal-directed coordination under severe temporal constraints.

Training Implications for Skill Acquisition

The present findings reinforce the effectiveness of established training methods designed to enhance interceptive performance, such as perceptual anticipation training (for reviews, Hadlow et al., 2018; Williams et al., 1999) and adaptive practice under temporal constraints (Gray, 2017, 2020), while also revealing aspects that warrant further development. By identifying the interceptive control mechanisms that limit batting performance under high temporal urgency, our results clarify why current approaches succeed and what additional components may improve them.

First, our findings help explain why training focused on the early extraction of anticipatory cues is effective. A defining characteristic of skilled performers is their ability to accurately predict future outcomes at an early stage by using kinematic and contextual information from their opponents (e.g., Abernethy, 1988; Costa et al., 2023; Jackson & Mogan, 2007; Jackson et al., 2020; Loffing & Hagemann, 2014; Müller et al., 2010; Paull & Glencross, 1997; for an overview, see Loffing & Cañal-Bruland, 2017), weighting and integrating these cues according to their reliability (e.g., Gray & Cañal-Bruland, 2018; Gredin et al., 2018, 2020; Harris et al., 2023; Helm et al., 2020; Mann et al., 2014; Nakamoto et al., 2022). Building on this evidence, perceptual anticipation training targets the same perceptual–cognitive processes and has demonstrated improvements in anticipatory skill (e.g., Burroughs, 1984) and transfer to sport performance (e.g., Broadbent et al., 2015). Baseball-relevant evidence demonstrates that visual anticipation skill is associated with actual batting performance (Morris-Binelli et al., 2018; Müller & Fadde, 2016), and that perceptual-cognitive training of pitch recognition can transfer to on-field batting performance (Fadde, 2016). Our results provide a mechanistic account of why such training might work. Under longer TTC conditions, batters maintained high contact rates and low timing errors, with early and well-differentiated swing trajectories and low variability near contact. As TTC shortened, divergence became delayed and compressed and variability increased. Together, these patterns suggest that accurate early prediction extends the available time for predictive motor planning, allowing the swing to be prestructured before initiation and reducing the need for late corrections. By strengthening the accuracy and reliability of priors, perceptual training expands the range of conditions under which predictive planning succeeds.

Second, our findings may help explain why training near the temporal limits of control is particularly effective for improving interceptive performance. In Gray's virtual batting studies (Gray, 2017, 2020), batters practiced at pitch speeds close to the threshold of successful contact. This temporal-threshold training produced the largest performance improvements in both virtual and real batting and was accompanied by stronger functional coupling between adjacent swing phases. Such coupling may indicate more effective online adjustments during swing execution. Whereas predictive control may dominate when sufficient time is available, performance near the success threshold may depend more critically on the ability to make fine-grained adjustments while the movement is unfolding. In this sense, the temporal boundary identified in the present study is consistent with the possibility that Gray's training enhanced the system's capacity for online coordination under severe time constraints.

Beyond explaining why existing training methods are effective, our findings point to new directions for developing approaches that could further enhance interceptive performance. One promising direction is to strengthen the capacity for online refinement and movement correction. Previous work has shown that skilled batters possess superior inhibitory control that enables them not only to stop inappropriate swings (Gray, 2009; Muraskin et al., 2015; Nakamoto & Mori, 2008) but also to reprogram ongoing movements in response to unexpected changes in the ball trajectory (Nakamoto et al., 2013; Nakamoto & Mori, 2012). Building on this, training should not merely emphasize early-cue anticipation but also create conditions that force late adjustments (Nasu et al., 2020). For example, by using early occlusion or pitch-tunneling paradigms—in which different pitch types share nearly identical early trajectories—batters would sometimes experience prediction errors that require midswing correction. Such practice would engage inhibitory, adaptive-timing, and reprogramming mechanisms that underpin flexible online control.

Finally, performance breakdown around 400 ms TTC, despite relatively stable perceptual sensitivity, highlights the importance of precise motor planning and command. Variability in swing trajectories under high time pressure indicates that perception remained accurate but that preplanned motor commands lacked sufficient precision to transform perception into effective action. Within predictive-processing and internal-model frameworks, such limitations arise when the mapping between predicted swing trajectories and motor output is imprecise. Each swing may provide an opportunity to recalibrate this mapping through comparison between the predicted and the observed outcomes of bat–ball contact (Krakauer, 2009; Shadmehr & Krakauer, 2008; Wolpert & Kawato, 1998). However, eye-tracking studies show that batters rarely maintain fixation on the contact point (Mann et al., 2013; Toole & Fogt, 2021), suggesting that visual feedback for evaluating contact accuracy is often limited. Even when predictive saccades bring gaze toward the expected contact location (e.g., Kishita et al., 2020a, 2020b; Mann et al., 2013), the duration of physical contact between bat and ball is extremely brief, on the order of 1 ms or less (Nicholls et al., 2006; Watts & Bahill, 1989), allowing little time for detailed visual analysis of impact outcomes. Training that provides clear postswing feedback on spatial and temporal contact error may therefore facilitate internal-model updating and improve the precision of preplanned motor control. Although our study did not directly measure motor planning, the behavioral patterns observed provide indirect evidence that the precision of preplanned control constrains performance when perceptual prediction is not the limiting factor.

Conclusion

The present study contributes to the long-standing debate on whether high-speed interception is controlled by preplanned or continuous online control. Our results indicate that both components were engaged under high temporal urgency and are consistent with a hybrid control framework. Pure online control was evident in the late divergence of trajectories and variability increases under urgency. Preplanned control was evident in the proactive advancement of swing initiation while preserving a consistent swing duration, early and stable divergence at long TTCs, and the dissociation between perceptual sensitivity and motor errors. Neither mechanism alone suffices. Instead, batting seems to rely on a hybrid system, where a prespecified motor plan provides a scaffold that is continuously refined by online corrections, until both mechanisms collapse under severe temporal constraints. This conclusion aligns with Gray's (2021) critique of the ballistic-only view and supports a broader shift toward recognizing the importance of online control in high-speed sport actions. At the same time, it acknowledges the enduring relevance of model-based, predictive planning. By situating our findings within optimal feedback control and predictive processing frameworks, we show that batting exemplifies general principles of motor control: the optimization of feedback gains, the minimization of prediction error, and the hierarchical integration of priors and evidence. Ultimately, our results suggest that theories of interception must move beyond binary contrasts between model-based and online accounts. To explain and predict how both prediction and online correction contribute to successful interception and specify how their balance shifts with task constraints, we suggest that integrating both theoretical accounts into hybrid models is more fruitful. Baseball batting, with its extreme temporal demands, provides an ideal testbed for developing and validating such theories.

Limitations on Generality

Several limitations must be acknowledged. First, the temporal threshold identified ($\sim$400 ms TTC) may not generalize across skill levels. Professional batters, who face higher pitch velocities in real games, may be able to sustain effective hybrid control under shorter TTCs. Thus, while our findings establish a boundary condition, the exact temporal value is likely skill-dependent. Second, uncertainty manipulation in this study was simplified. We provided explicit information about the distribution of pitch locations, which may have reduced the ecological impact of uncertainty. In real competition, contextual priors are probabilistic and less reliable. Bayesian integration models suggest that interceptive performance depends critically on the relative precision of priors and sensory evidence (Adams et al., 2013; Brantley & Körding, 2022; Körding & Wolpert, 2004). Future work should vary the reliability of contextual priors to better capture real-world conditions. Related to this broader simplification of uncertainty, the informational context of the present task necessarily simplified several aspects of batting. In the present study, all pitches were delivered by the same avatar with identical pitching kinematics, and interaction with a live pitcher was not included. This design limited variability in pitch characteristics and removed rich sources of contextual information, such as situational probabilities based on pitch count or pitcher tendencies, that typically inform anticipatory behavior in real-game situations. Because predictive control relies on the reliability of contextual and kinematic cues, constraining these sources of information may have reduced the relative weight assigned to advance predictions. Under such informational constraints, the balance of hybrid control may therefore have been biased toward greater reliance on online adjustments during swing execution than would be expected in more diverse and information-rich environments that better support predictive control. At the same time, the inclusion of multiple pitch types, particularly those involving late trajectory changes, would be expected to alter cue reliability in a different way. For example, when late-breaking pitches are intermixed with fastballs, early ball-flight and kinematic cues may become less diagnostic of the final trajectory, thereby increasing the functional importance of within-swing online regulation. Together, these considerations suggest that the relative contribution of predictive and online components may vary systematically depending on how reliable and diagnostic the available cues are within a given task context. Future studies that systematically vary pitch characteristics and cue availability will be important for examining how this balance changes across different interceptive environments.

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